[Aim : 100/100 in Maths]

आरंभ

REAL NUMBERS

Lecture #3

ATK^2 :-

→ HCF , LCM → prime factorisation.

→ Prime no. → only two factors < 1, itself {2, 3, 5, 7, 11 --}

Composite no. → more than 2 factors {4, 6, 8, 9, 10 - -}

K^3B : Prime no.s के Rebted :

$5 = 5^1$

$7 = 7^1$

HCF (5, 7) = 1

LCM (5, 7) = $5^1 \times 7^1$

= 35

$11 = 11^1$

$13 = 13^1$

HCF (11, 13) = 1

LCM (11, 13) = $11^1 \times 13^1$

= 143

⊛ For two prime no.s (p & q)

- HCF = 1 (fix)
- LCM = product of two no.

$\boxed{LCM = p \times q}$

Lallu Problem

#LP: LCM of two prime numbers p and q (p>q) is 221. Find the value of 3p - q :

(A) 2

(B) 28

(C) 38 ✓

(D) 48

$LCM = p \times q$

ATQ → $221 = p \times q$

$13 \times 17 = p \times q$

given $p > q$

$\therefore$ on comparing $p = 17$, $q = 13$

$$\begin{array}{r|l} 13 & 221 \\ \hline 17 & 17 \\ \hline & 1 \end{array}$$

Final Ans → $3p - q$

$\Rightarrow 3(17) - 13$

$\Rightarrow 51 - 13$

$\Rightarrow 38$

#LP: The LCM of smallest 2-digit number and smallest composite number is: [CBSE 2023]

(A) 12

(B) 4

(C) 20 ✓

(D) 40

10 4

$LCM(4, 10) = \underline{20}$

#LP: If 'p' and 'q' are natural numbers and 'p' is the multiple of 'q', then what is the HCF of 'p' and 'q'? [CBSE 2023]

(A) pq
(B) p
(C) q ✓
(D) p + q

p is multiple of q.

random example → let, $q = 2$, $p = 8$ } Natural No ✓

$2 = 2$
$8 = 2 \times 2 \times 2$
HCF = 2

HCF (2, 8) ⇒ 2 = q

#LP: HCF of two consecutive natural numbers is?

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random example

Let the no.s are → 8, 9

$8 = 2 \times 2 \times 2 \times 1$

$9 = 3 \times 3 \times 1$

$\boxed{HCF = 1}$

$\boxed{HCF(8, 9) = 1}$

Ans

Next Toppers

#LP: Assertion (A): HCF of any two consecutive even natural numbers is always 2. True

Reason (R): Even natural numbers are divisible by 2. True [CBSE SQP 2024-25]

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Reason:- Even NN → 2, 4, 6, 8, 10

div. by 2

Common factor is 2

Even NN ⇒ $\underbrace{2, 4}_{\text{conseq.}}$, 6, $\overline{8, 10}$ - - -

HCF (2, 4) = 2

HCF (8, 10) = 2 ✓

K^3B :-

★ HCF is always factor of LCM.

$\frac{LCM}{HCF}$ divide कर देगा

★ $HCF < LCM$

#LP: The LCM of two numbers is 2400. Which of the following CANNOT be their HCF? [CBSE 2022]

(A) 300
(B) 400
(C) 500
(D) 600

LCM = 2400

HCF is factor of LCM

(A) $\frac{2400}{300} = 8$ ✓

(B) $\frac{2400}{400} = 6$ ✓

(C) $\frac{2400}{500}$ ✗

$5 \overline{)24}(4$
20
4 → Rem.

HW

#LP: Assertion (A) : H.C.F. $(36m^2, 18m) = 18m$, where m is a prime number.

Reason (R) : H.C.F. of two numbers is always less than or equal to the smaller number. [CBSE 2026]

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

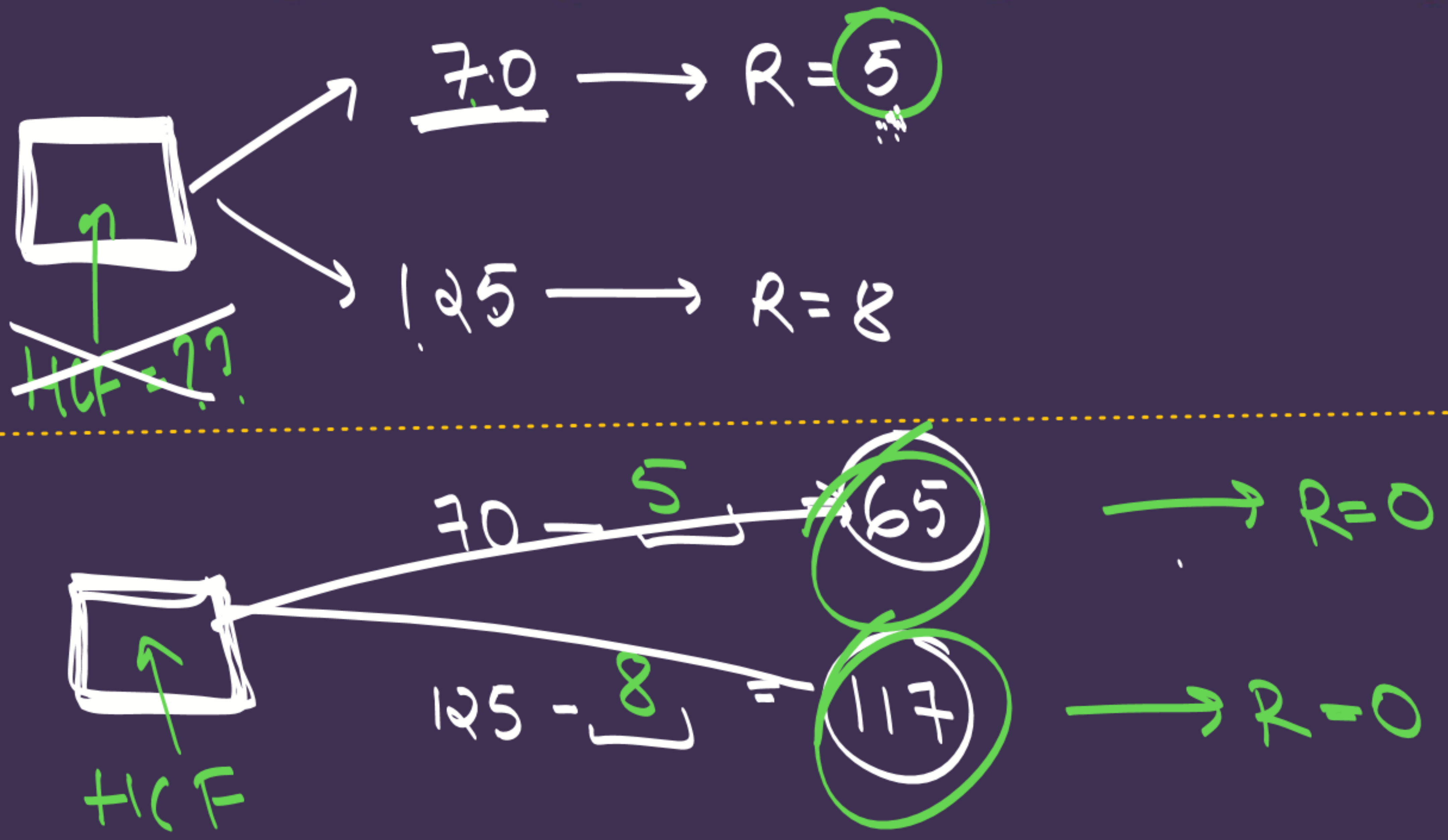
#LP: The largest number which divides 70 and 125, leaving remainders 5 and 8, respectively, is [NCERT Examplar]

(A) 13

(B) 65

(C) 875

(D) 1750



final Ans → $\underline{HCF(65, 117)} = 13$ Ans

$$65 = 5 \times 13$$
$$117 = 3 \times 3 \times 13$$

$$HCF = 13$$

HCF → वो सबसे बड़ा No. जो दोनों को completely divide कर दे

$\boxed{R = 0}$

$2\overline{)7}(3$
6
1 ← $R = 1$

$\boxed{7 - R}$

$7 - 1 \Rightarrow 6$

completely divide

$2\overline{)6}(3$
6
0 ← $R = 0$

#LP: Using Euclid's division algorithm, find the largest number that divides 1251, 9377 and 15628 leaving remainders 1, 2 and 3, respectively.

y

1251 ⟶ R = 1

9377 ⟶ R = 2

15628 ⟶ R = 3

HCF

1251 − 1 → 1250 — R = 0

9377 − 2 → 9375 — R = 0

15628 − 3 → 15625 — R = 0

HCF (1250, 9375, 15625)

= 625

2	1250
5	625
5	125
5	25
5	5
	1

3	9375
5	3125
5	625
5	125
5	25
5	5
	1

5	15625
5	3125
5	625
5	125
5	25
5	5
	1

$$1250 = 2 \times 5 \times 5 \times 5 \times 5$$
$$9375 = 3 \times 5 \times 5 \times 5 \times 5 \times 5$$
$$15625 = 5 \times 5 \times 5 \times 5 \times 5 \times 5$$

$$HCF = 5 \times 5 \times 5 \times 5$$
$$= \underline{\underline{625}}$$ Ans

Module → 1 P I → 1, 5,
1 P II → 3
2 P I → 6, 7
2 P II → 1, 4, 5
2 P III → 6

THANK YOU COODIESS !!

#LP 1 : If HCF of $84x^2$ and $126x^3$ is 42, then find the value of x.

#LP 2: Let a and b be two positive integers such that $a = p^4q^2$ and $b = p^3q^5$, where p and q are prime numbers. If HCF(a, b) = p^mq^n and LCM(a, b) = p^rq^s, then (m + n) + (r + s) =

(A) 12
(B) 14
(C) 16
(D) 18

#LP 3: If two positive integers a and b are written as $a = x^4y^3$ and $b = x^2y^5$, x and y are prime numbers, then HCF (a, b) is

(A) x^2y^3
(B) x^4y^5
(C) x^2y^5
(D) x^4y^3

#LP 4: If the HCF of 210 and 55 is expressible in the form 210 × 5 + 55y, then find the value of y.

(A) -19
(B) 19
(C) 5
(D) -5

#LP 5: The ratio of HCF to LCM of the least composite number and the smallest even prime number is:

(A) 1:1
(B) 1:2
(C) 2:1
(D) 1:4

#LP 6: The natural number 0 is:

(A) a prime number

(B) a composite number

(C) both prime and composite

(D) neither prime nor composite

#LP 7: If $x = a^2b^3$ and $y = a^4b$, where a and b are prime numbers, then [HCF(x, y) – LCM(x, y)] is: [CBSE Pattern]

(A) $a^2b - a^4b^3$

(B) $ab\ (1 - a^3b^2)$

(C) $a^2b + a^4b^3$

(D) $ab\ (1 - ab)(1 + ab)$

#LP 8: The HCF of $2^6 \cdot 5^4$ and $2^3 \cdot 5^7$ is:

(A) 1

(B) $2^3 \cdot 5^4$

(C) $2^6 \cdot 5^7$

(D) $2^3 \cdot 5^3$